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ISGF
India Smart Grid Forum

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

Supporting Ministries

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CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF
SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

Session : DIGITALIZATION ROADMAPS FOR UTILITIES

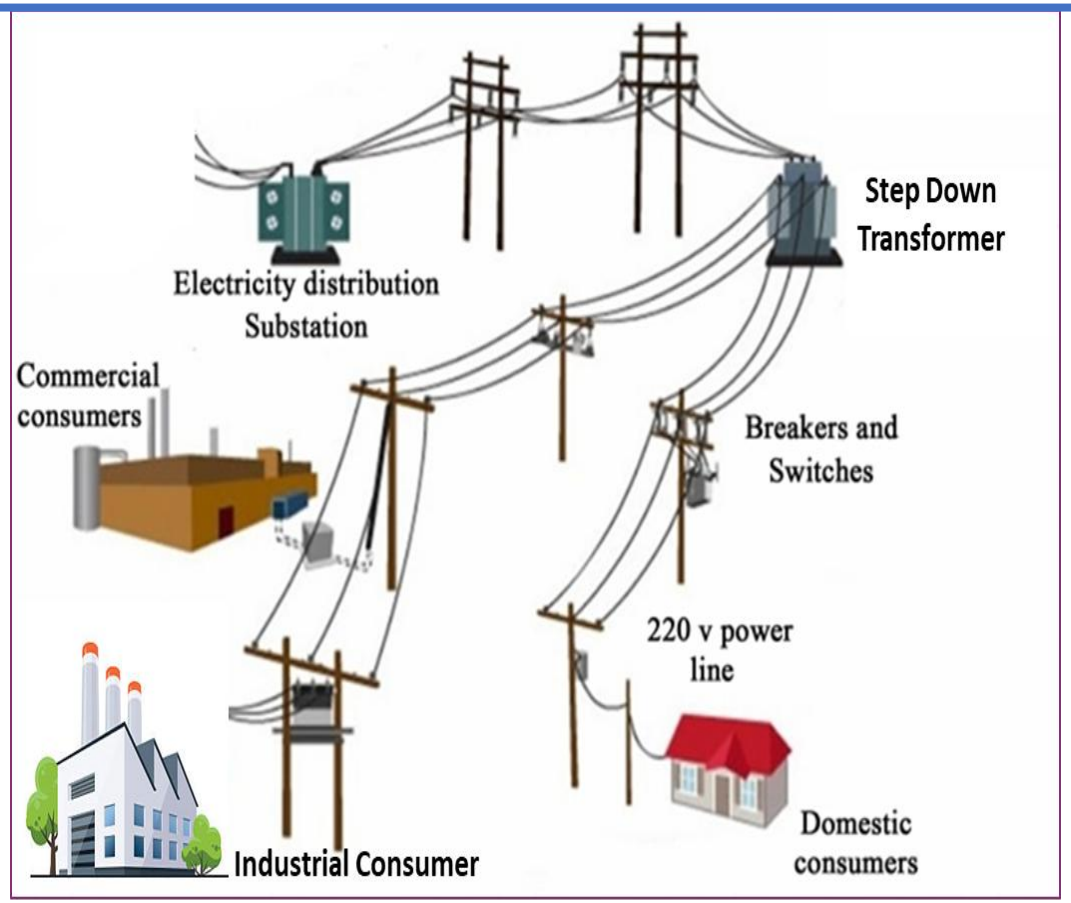
PRESENTATION TOPIC

Presented By

NOIDA POWER COMPANY LIMITED

Company Profile

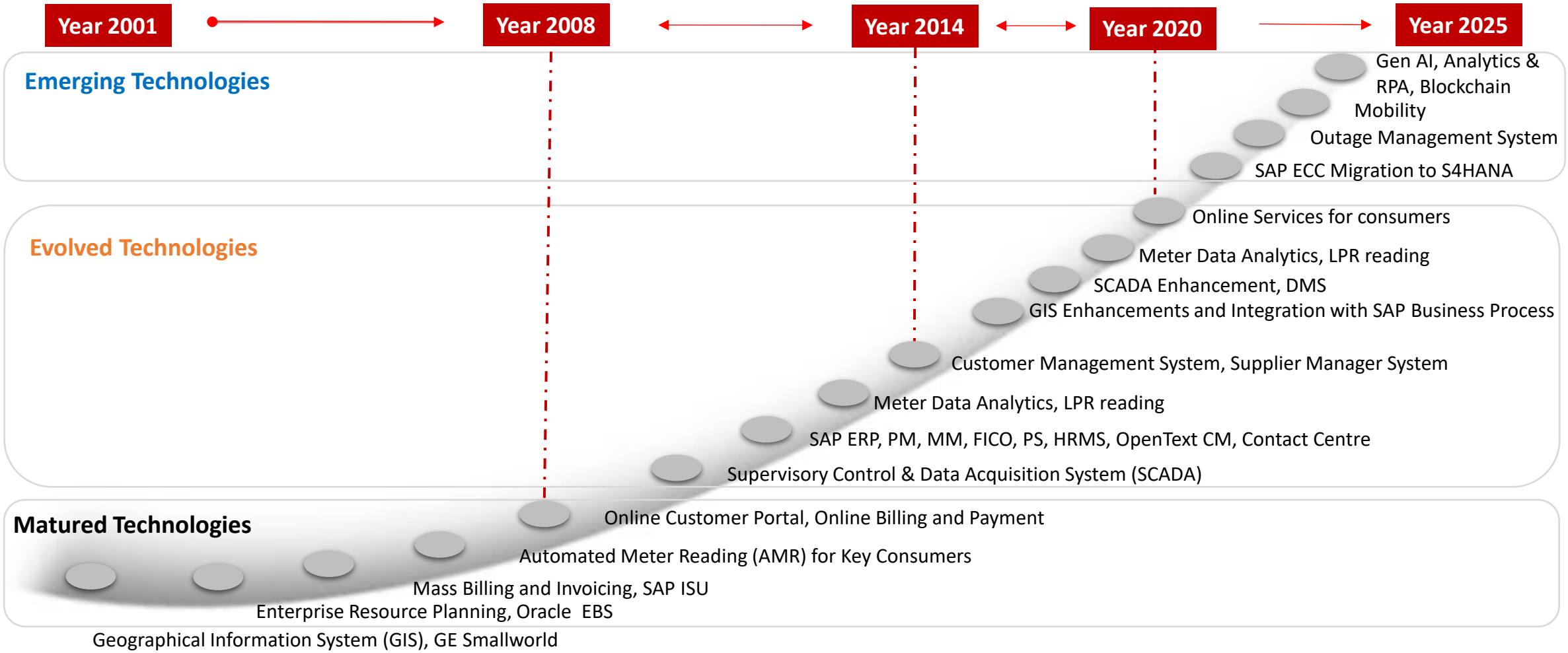
- Joint venture between RP-Sanjiv Goenka group and GNIDA
- Licensed Area : Greater Noida (335 km²)
- Population catered: 1 Million +
- Average Supply Restoration Time : 1.34 hours
- Sales Mix: HT | LT, 70% | 30%
- Billing Accuracy: 99.9% | Meter Reading Efficiency: 98%
- Collection Efficiency: 100.34%
- Overall Compliance to Complaints: 97.2%



Certifications

ISO 9001:2015 (QMS)	ISO 50001:2018 (EnMS)	ISO 45001:2018 (OH&SMS)	ISO 27001:2013 (ISMS)	ISO 14001:2015 (EMS)	ISO 10002:2018 (CSAT)

Digitalization Journey at NPCL



Technological innovation has been a regular feature

Digital Adoption at NPCL

One of the largest IT Setup in Greater Noida

- Large Fiber Network covering Length and Breadth of Greater Noida
- More than 65 Offices connected
- World Class Data Center equipped with latest equipment
- More than 150 Virtual Servers
- Nearly 4000 IT Assets - PC's / Printers / UPS / Switches etc.

Best in Class Applications

- SAP ERP, ISU and CRM
- GE Open Office as GIS for Network Management
- ETAP SCADA
- Low Code NO Code Platform – APEX Applications
- Blue Prism as Robotics Process Automation Tool
- Qlik as Business Intelligence Tool and Dashboards
- Python / R for Artificial Intelligence & Data Analytics
- Gen AI for Video / Images / Documents Analytics

Framework for Digital Adoption at NPCL

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Assess Digital Maturity

- Metering systems
- SCADA / grid monitoring
- IT and billing systems
- Communication networks
- Cybersecurity readiness

1

Define Strategic Objectives

- reduction of AT&C losses
- improved reliability and outage management
- Renewable integration
- improved consumer experience

2

Identify Priority Digital Platforms

- Advanced Metering Infrastructure (AMI)
- Meter Data Management System (MDMS)
- SCADA / DMS
- Outage Management System (OMS)
- GIS-based asset management

3

Phased Implementation Strategy

- Phase 1 — Digital Foundation
- Phase 2 — Grid Intelligence
- Phase 3 — Intelligent Utility

4

Implemented Technology & Systems

Front office & consumer interaction channels

- Social media
- Call centre
- Portal
- ChatBot
- Consumer App
- Mobile wallet & payment gateways

Mobile Apps to workforce

- On Line Metering Reading
- Commercial & Technical feasibility on field
- Surveillance
- Safety
- Shift Engineer
- Crew
- Proof of Bill Delivery
- Online Meter Management

Systems for strong foundation

- OMS
- Grid & DT AMR
- Consumer AMR
- SCADA, DMS
- GIS
- SAP CRM
- SAP FI
- SAP PS
- SAP MM, SRM
- SAP PM
- SAP SD
- SAP HCM
- SAP ISU-CS, CM
- SAP ISU-DM
- SAP ISU-Billing

AI / ML / Gen AI / Agentic AI

Cyber Security

New Technologies

Integrated IT & OT Landscape

NPCL IT Journey

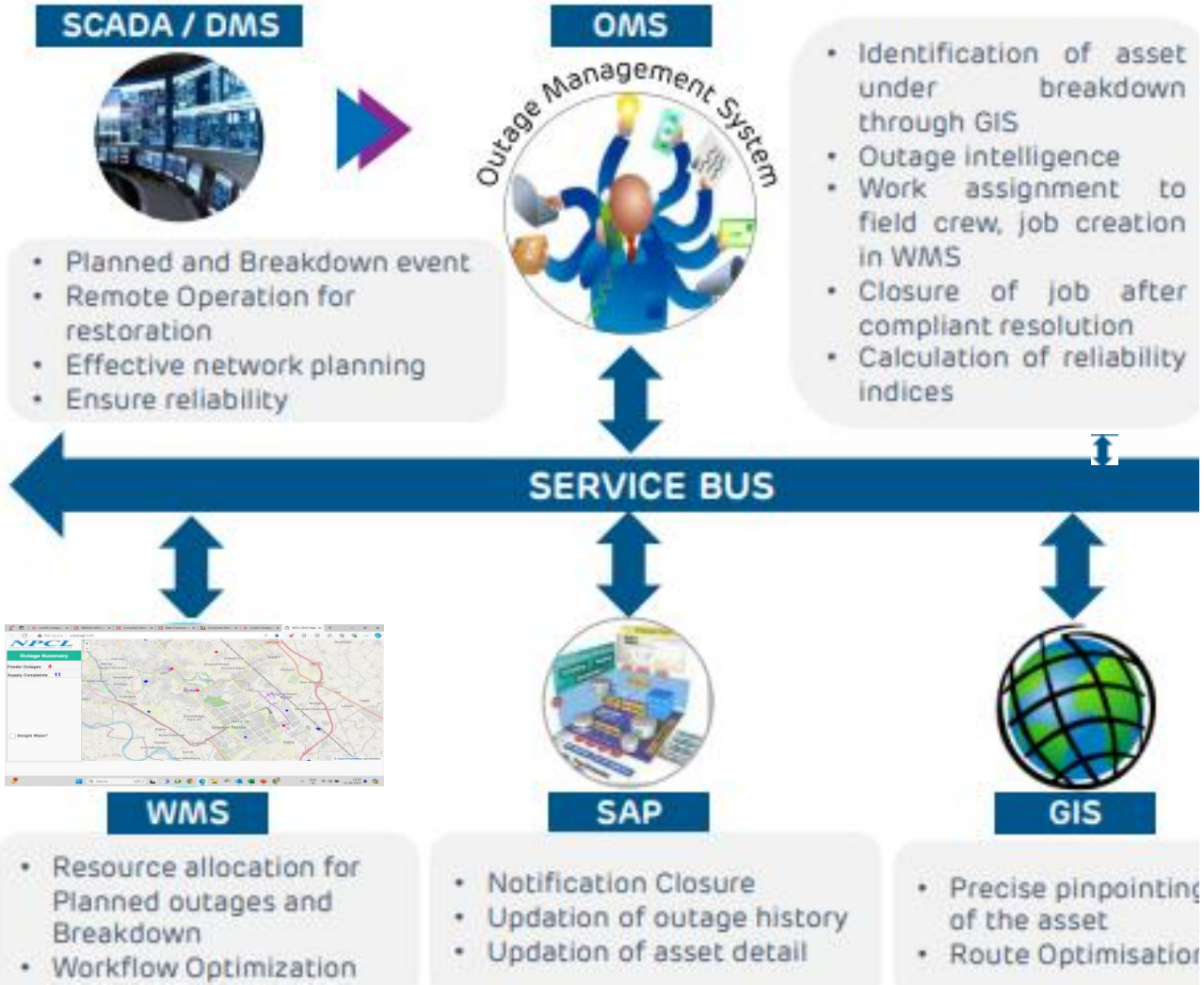
- Implemented Technology & Systems
- AI Landscape
- Analytics Initiatives
- Cyber Security Readiness
- OT Asset Management
- Implementation of Drone Technology
- Cyber Security Readiness
- Virtual Reality (VR) Implementation

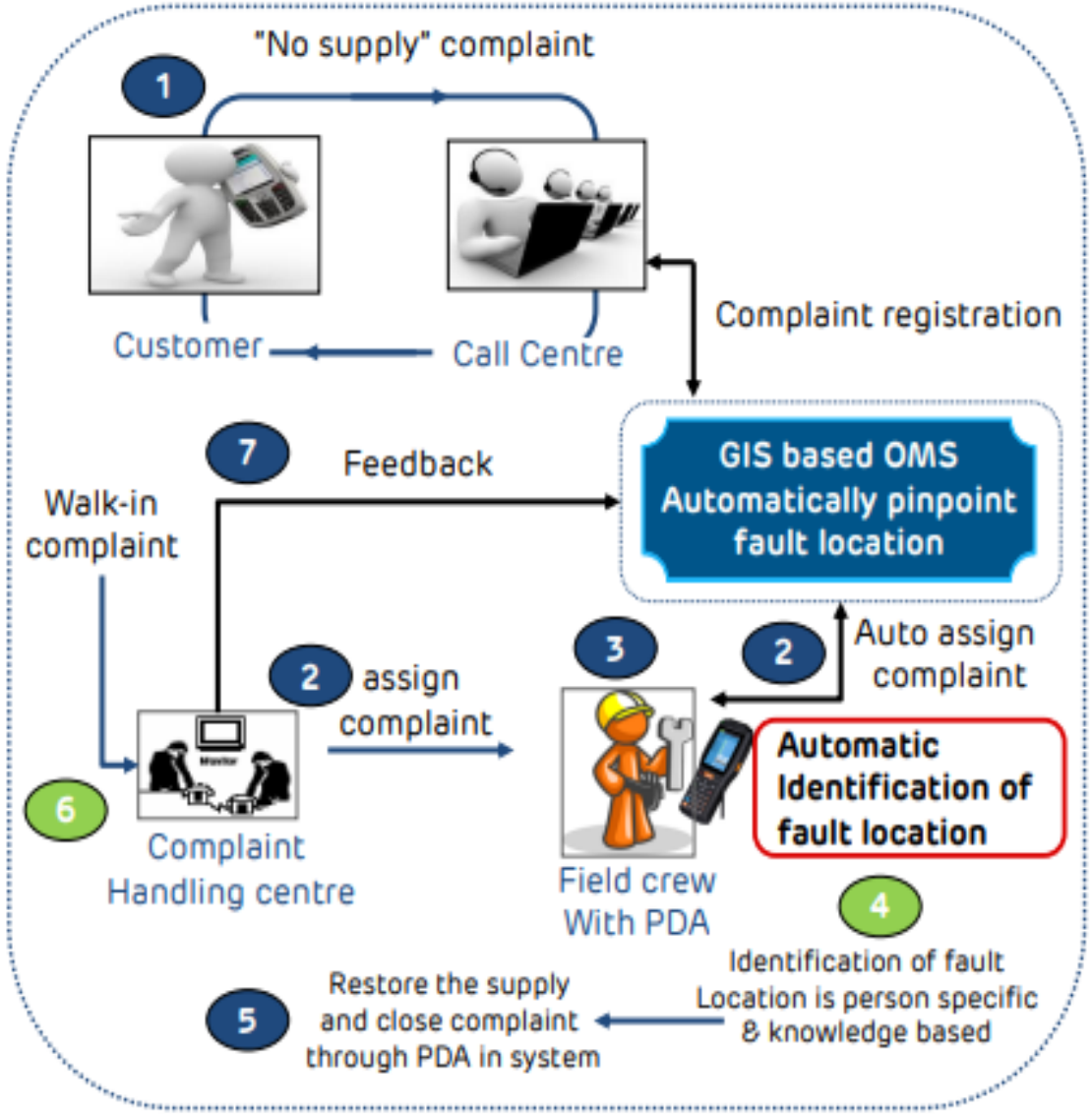
IT & OT Integration

NPCL OT Journey

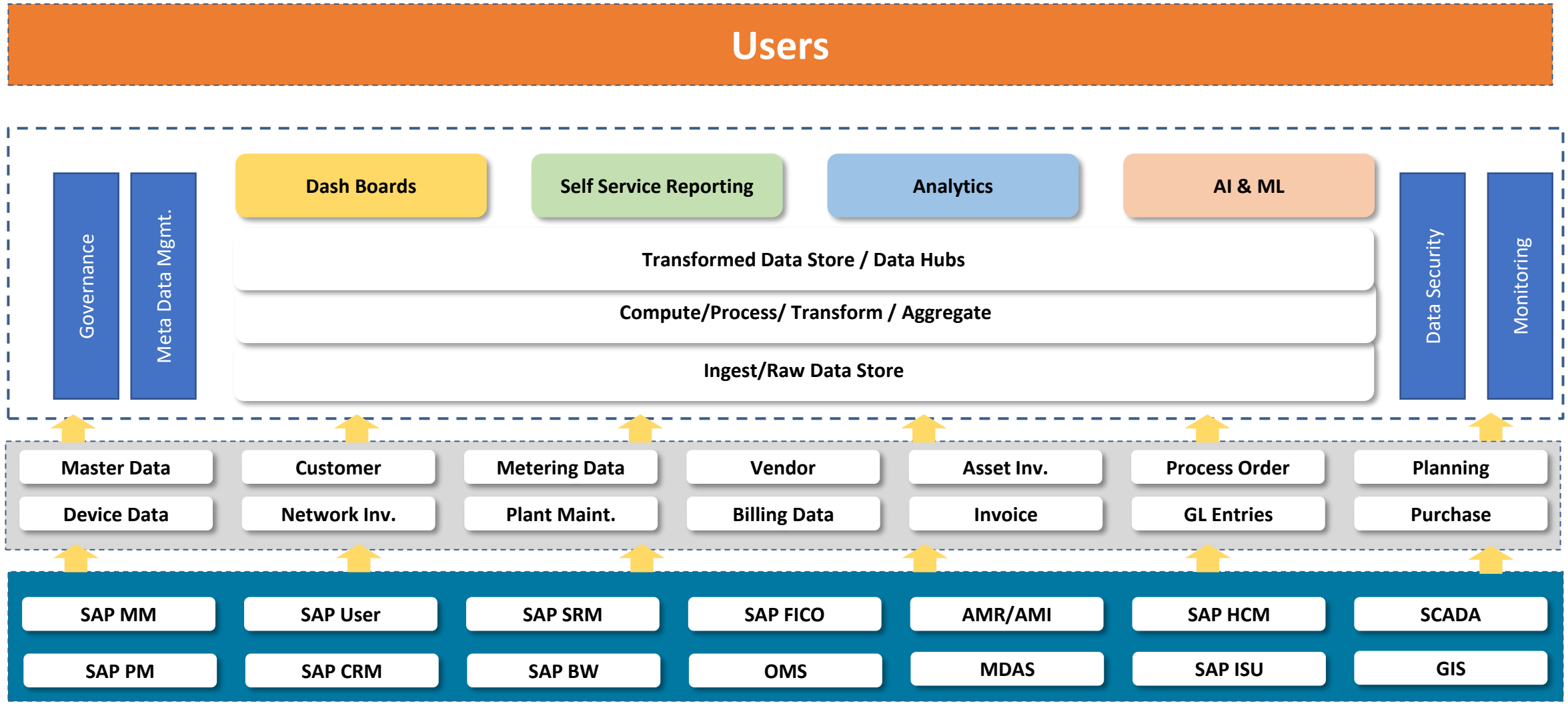
- SCADA enabled 24/7 Centralized Controlled Operations
- GIS and SAP Integration Business Processes
- Integrated Inhouse Outage Management System (OMS)
- Distribution Management System (DMS)
- Distribution Transformer Health Monitoring
- Self Healing Grid Mechanism
- Enhanced Safety through Technology Inclusion
- SMART Substations
- Battery Health Monitoring System
- SCADA & SMART GRID Lab

Integration Application Roadmap





Integrated AI Landscape



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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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